

Mariam Modular Runtime Object Model

Version v130 draft

README.md

Volume 00 - Enterprise Constitution

Version: v1 draft

Status: Draft

Document ID: MAL-V00-README

Purpose

Volume 00 establishes the constitutional foundation for Mariam AI Enterprise OS. It defines the durable identity, principles, operating expectations, governance model, quality posture, security posture, engineering behavior, lifecycle model, and long-term direction that all later volumes must respect.

Scope

This volume applies to every business domain, AI capability, product module, integration, contributor workflow, and release decision in Mariam AI Enterprise OS.

Principles

- Constitutional rules are stable enough to guide multiple product releases.
- Every downstream volume must remain compatible with this foundation.
- Governance, quality, security, and engineering discipline are treated as product foundations.

Operating Rules

- Update this volume only through explicit versioned changes.
- Trace roadmap, blueprint, specification, and testing decisions back to constitutional documents.
- Keep constitutional language implementation-independent unless a binding constraint is intentional.

Documents

- INDEX.md
- CHANGELOG.md
- TRACEABILITY.md
- ACCEPTANCE.md
- book-00-01-identity.md through book-00-15-long-term-strategy.md

Acceptance Criteria

- Every constitutional book has a stable document ID.
- Every implementation roadmap item can trace back to one or more constitutional principles.
- Changes to constitutional rules are versioned, reviewed, and recorded.

Version History

Version	Date	Status	Notes
v1 draft	2026-06-29	Draft	Initial Volume 00 structure and content.

■# Volume 00 Index

Version: v1 draft

Status: Draft

Document ID: MAL-V00-INDEX

Purpose

This index defines the reading order and canonical location of every Volume 00 document.

Scope

The index covers constitutional material only and does not replace roadmap, blueprint, or specification indexes.

Principles

- Keep document names stable after publication.
- Add new constitutional books only when they govern broad enterprise behavior.
- Preserve numeric order to make references durable.

Operating Rules

- A document listed here is considered part of the constitutional baseline.
- Deprecated documents remain listed until an approved replacement exists.
- Cross-volume references must use document IDs, not informal names.

Acceptance Criteria

- All Volume 00 files are listed.
- Each listed item exists in the repository.
- The index is updated before any Volume 00 release.

Canonical Books

1. MAL-V00-B001 - Identity
2. MAL-V00-B002 - Vision
3. MAL-V00-B003 - Mission
4. MAL-V00-B004 - Core Principles
5. MAL-V00-B005 - Enterprise Principles
6. MAL-V00-B006 - AI Principles
7. MAL-V00-B007 - DNA Principles
8. MAL-V00-B008 - Swarm Principles
9. MAL-V00-B009 - Governance Principles
10. MAL-V00-B010 - Quality Principles
11. MAL-V00-B011 - Security Principles

- 12. MAL-V00-B012 - Engineering Principles
- 13. MAL-V00-B013 - Evolution Principles
- 14. MAL-V00-B014 - Development Lifecycle
- 15. MAL-V00-B015 - Long-Term Strategy

Version History

Version	Date	Status	Notes
v1 draft	2026-06-29	Draft	Initial index.

- MAL-V00-B16 - ■ook-00-16-modular-changeability-principle.md - Modular Changeability Principle

■# Volume 00 Changelog

Version: v1 draft

Status: Draft

Document ID: MAL-V00-CHANGELOG

Purpose

This changelog records material changes to the constitutional baseline.

Scope

It covers changes to Volume 00 content, document status, document IDs, acceptance rules, and traceability rules.

Principles

- Constitutional history must remain auditable.
- Changes must explain why the library evolved.
- Draft changes are recorded with the same seriousness as approved changes.

Operating Rules

- Every release adds an entry.
- Entries identify affected documents and governance impact.
- Breaking constitutional changes require explicit migration notes.

Acceptance Criteria

- The changelog includes the current release.
- Each entry states date, version, status, and summary.
- No release artifact is produced without a changelog update.

Entries

Date	Version	Status	Summary
2026-06-29	v1 draft	Draft	Created the initial Mariam Enterprise Constitution and release framework.

Version History

Version	Date	Status	Notes
v1 draft	2026-06-29	Draft	Initial changelog.

v130 Entries

Date	Version	Status	Summary
2026-06-30	v130 draft	Draft	Added Modular Changeability Principle.

■# Volume 00 Traceability

Version: v1 draft

Status: Draft

Document ID: MAL-V00-TRACEABILITY

Purpose

Traceability makes constitutional authority visible from principle to roadmap, specification, implementation, test, and release.

Scope

This document applies to all Volume 00 principles and all downstream documents that depend on them.

Principles

- Every major decision must be traceable to a constitutional source.
- Trace links must be stable enough for release review.
- Absence of traceability is treated as unresolved design debt.

Operating Rules

- Use document IDs in roadmap, blueprint, specification, and testing references.
- Link requirements to one or more constitutional principles when they affect system behavior.
- Record unresolved trace gaps before release approval.

Acceptance Criteria

- Each Volume 00 book has a document ID.
- Release notes identify included constitutional documents.
- Future implementation issues can reference constitutional IDs.

Trace Map

Source	Downstream Use
MAL-V00-B001 to MAL-V00-B003	Product identity, executive messaging, contributor alignment
MAL-V00-B004 to MAL-V00-B009	Governance, roadmap approval, architectural decisions
MAL-V00-B010 to MAL-V00-B012	Engineering standards, testing policy, security controls
MAL-V00-B013 to MAL-V00-B015	Lifecycle planning, release evolution, strategic expansion

Version History

Version	Date	Status	Notes
v1 draft	2026-06-29	Draft	Initial traceability model.

v130 Traceability

Document	Traceability
MAL-V00-B16	Governs add, edit, delete, disable, enable, upgrade, replace, fork, rollback, DNA export, and DNA import for all Mariam objects.

■# Volume 00 Acceptance

Version: v1 draft

Status: Draft

Document ID: MAL-V00-ACCEPTANCE

Purpose

This document defines the acceptance rules for treating Volume 00 as a usable constitutional baseline.

Scope

It covers document completeness, review readiness, release packaging, and readiness for downstream volumes.

Principles

- A constitutional release must be coherent before it is comprehensive.
- Acceptance is based on usable governance value, not page count.
- Draft status still requires internal consistency.

Operating Rules

- Each book must include purpose, scope, principles, operating rules, acceptance criteria, and version history.
- Release artifacts must include Markdown, PDF, ZIP, manifest, and release notes.
- Missing downstream volumes are acceptable only when clearly reserved.

Acceptance Criteria

- Volume 00 contains README, INDEX, CHANGELOG, TRACEABILITY, ACCEPTANCE, and 15 books.
- Generated PDF includes every Volume 00 Markdown file in deterministic order.
- Manifest records all generated release artifacts with size and checksum.

Version History

Version	Date	Status	Notes
v1 draft	2026-06-29	Draft	Initial acceptance criteria.

Modular Changeability Principle Acceptance

- Every governed Mariam object is treated as lifecycle-managed and changeable through approved operations.
- Governance, permission, versioning, audit, compatibility, testing, and rollback are mandatory.

Book 00-01 - Identity

Version: v1 draft

Status: Draft

Document ID: MAL-V00-B001

Purpose

Define Mariam AI Enterprise OS as a governed enterprise operating system for AI-enabled work, not a loose collection of tools.

Scope

Covers naming, identity boundaries, product posture, and the relationship between documentation and implementation.

Principles

- Mariam is an enterprise OS for orchestrating AI, human work, knowledge, governance, and execution.
- The system identity is durable even when modules, vendors, or implementation technologies change.
- Documentation is part of the product, not an accessory.

Operating Rules

- Use the full name in constitutional and release contexts.
- Do not redefine Mariam as a chatbot, single app, marketplace only, or automation script.
- Reject features that weaken the enterprise operating-system identity.

Acceptance Criteria

- A contributor can state what Mariam is in one paragraph.
- The repository README identifies this library as the Source of Truth.
- Future volumes can reference identity without ambiguity.

Version History

Version	Date	Status	Notes
v1 draft	2026-06-29	Draft	Initial canonical draft for the Mariam Architecture Library.

Book 00-02 - Vision

Version: v1 draft

Status: Draft

Document ID: MAL-V00-B002

Purpose

State the future Mariam is designed to make possible for organizations and independent workers.

Scope

Covers strategic aspiration, product horizon, and the intended transformation of AI-enabled work.

Principles

- Mariam should make complex work governable, inspectable, and continuously improvable.
- AI agents must strengthen human capability instead of hiding decisions.
- The long-term product should connect work, knowledge, markets, and operational memory.

Operating Rules

- Evaluate roadmap items by contribution to the vision.
- Prefer capabilities that compound across domains.
- Avoid short-term convenience that creates long-term opacity.

Acceptance Criteria

- The vision can guide roadmap prioritization.
- The vision is broad enough for multiple domains but concrete enough to constrain design.
- Future expansion can be judged against this document.

Version History

Version	Date	Status	Notes
v1 draft	2026-06-29	Draft	Initial canonical draft for the Mariam Architecture Library.

Book 00-03 - Mission

Version: v1 draft

Status: Draft

Document ID: MAL-V00-B003

Purpose

Define the operational mission that turns the vision into repeatable execution.

Scope

Covers what Mariam must reliably do for users, teams, contributors, and operators.

Principles

- Convert goals into governed workflows.
- Preserve knowledge produced during work.
- Coordinate AI and human contributors with accountability.

Operating Rules

- Mission-critical workflows require explicit ownership and traceability.
- Outputs must be inspectable before they become authoritative.
- The system must support real operating environments, not demo-only paths.

Acceptance Criteria

- Mission statements map to product objectives in Volume 01.
- Core workflows can be tested against mission outcomes.
- Operational success can be measured without relying on vague adoption claims.

Version History

Version	Date	Status	Notes
v1 draft	2026-06-29	Draft	Initial canonical draft for the Mariam Architecture Library.

Book 00-04 - Core Principles

Version: v1 draft

Status: Draft

Document ID: MAL-V00-B004

Purpose

Declare the cross-cutting principles that govern every Mariam decision.

Scope

Covers simplicity, explicitness, traceability, modularity, accountability, and user trust.

Principles

- Make important state visible.
- Design modules with clear boundaries.
- Prefer boring reliability over impressive fragility.
- Treat governance as a user experience.

Operating Rules

- No critical workflow may depend on hidden assumptions.
- Every module must have a defined responsibility.
- Complexity must earn its place through measurable value.

Acceptance Criteria

- Architecture reviews can cite core principles.
- Unclear ownership is flagged before implementation.
- Product behavior remains understandable to operators.

Version History

Version	Date	Status	Notes
v1 draft	2026-06-29	Draft	Initial canonical draft for the Mariam Architecture Library.

Book 00-05 - Enterprise Principles

Version: v1 draft

Status: Draft

Document ID: MAL-V00-B005

Purpose

Define how Mariam behaves as enterprise software for teams, organizations, and managed operations.

Scope

Covers roles, tenancy, auditability, policy, scalability, and operational durability.

Principles

- Enterprise users need control, audit, reliability, and predictable change.
- Policy must be enforceable across modules.
- Tenant and role boundaries must be explicit.

Operating Rules

- Design enterprise workflows around permissions, records, and review states.
- Do not add shared capabilities without ownership and audit semantics.
- Treat operational reporting as a primary requirement.

Acceptance Criteria

- Enterprise objectives in Volume 01 map to these principles.
- Future blueprints include role and audit considerations.
- Release readiness includes operational governance checks.

Version History

Version	Date	Status	Notes
v1 draft	2026-06-29	Draft	Initial canonical draft for the Mariam Architecture Library.

Book 00-06 - AI Principles

Version: v1 draft

Status: Draft

Document ID: MAL-V00-B006

Purpose

Define the constitutional rules for AI behavior inside Mariam.

Scope

Covers agent boundaries, explainability, safety, human control, evaluation, and model independence.

Principles

- AI must be assistive, inspectable, bounded, and accountable.
- Model providers are replaceable implementation details.
- High-impact AI actions require review, logging, and rollback paths.

Operating Rules

- Do not let AI silently mutate authoritative records.
- Capture prompts, context, decisions, and outputs where governance requires it.
- Separate recommendation, decision, and execution states.

Acceptance Criteria

- AI strategy in Volume 01 traces to this book.
- AI features define failure modes and review paths.
- Model changes can be evaluated without rewriting product identity.

Version History

Version	Date	Status	Notes
v1 draft	2026-06-29	Draft	Initial canonical draft for the Mariam Architecture Library.

Book 00-07 - DNA Principles

Version: v1 draft

Status: Draft

Document ID: MAL-V00-B007

Purpose

Define Mariam DNA as the persistent organizational memory and behavioral fingerprint of work.

Scope

Covers identity memory, preference capture, operating patterns, reusable knowledge, and adaptation.

Principles

- DNA is structured memory that improves execution over time.
- DNA must be consented, scoped, and correctable.
- DNA should encode patterns, not private noise.

Operating Rules

- Separate user DNA, team DNA, domain DNA, and system DNA.
- Make DNA updates reviewable when they affect future recommendations.
- Do not treat inferred memory as unquestionable fact.

Acceptance Criteria

- DNA strategy in Volume 01 can use this as its authority.
- Memory features include correction and provenance.
- Personalization never overrides security or governance.

Version History

Version	Date	Status	Notes
v1 draft	2026-06-29	Draft	Initial canonical draft for the Mariam Architecture Library.

Book 00-08 - Swarm Principles

Version: v1 draft

Status: Draft

Document ID: MAL-V00-B008

Purpose

Define how multiple agents, services, and contributors coordinate within Mariam.

Scope

Covers agent collaboration, task decomposition, consensus, escalation, and conflict handling.

Principles

- Swarm behavior must be orchestrated, not chaotic.
- Specialized agents require clear contracts.
- Conflicts should surface as reviewable events.

Operating Rules

- Assign every agent a bounded role and output contract.
- Record handoffs between agents when work affects authoritative outputs.
- Escalate unresolved contradictions to human review.

Acceptance Criteria

- Swarm workflows can be tested with deterministic scenarios.
- Agent outputs can be traced to source context.
- Coordination failures are observable.

Version History

Version	Date	Status	Notes
v1 draft	2026-06-29	Draft	Initial canonical draft for the Mariam Architecture Library.

Book 00-09 - Governance Principles

Version: v1 draft

Status: Draft

Document ID: MAL-V00-B009

Purpose

Define decision authority, review expectations, and change control for the enterprise OS.

Scope

Covers policy, approval, ownership, exceptions, and documentation authority.

Principles

- Governance should enable confident speed.
- Authority must be explicit.
- Exceptions expire unless renewed.

Operating Rules

- Assign owners to constitutional, roadmap, blueprint, and specification decisions.
- Document policy exceptions with rationale and review date.
- Treat undocumented decisions as provisional.

Acceptance Criteria

- Release notes identify governance impact.
- Roadmap changes can cite approval authority.
- Contributors know where decisions belong.

Version History

Version	Date	Status	Notes
v1 draft	2026-06-29	Draft	Initial canonical draft for the Mariam Architecture Library.

Book 00-10 - Quality Principles

Version: v1 draft

Status: Draft

Document ID: MAL-V00-B010

Purpose

Define what quality means for Mariam before implementation details exist.

Scope

Covers correctness, usability, resilience, observability, accessibility, and maintainability.

Principles

- Quality is measured across product behavior, documentation, operations, and user trust.
- A workflow is not complete until failure behavior is designed.
- Usability and reliability are architectural concerns.

Operating Rules

- Each specification must define success and failure behavior.
- Testing must cover permissions, state transitions, and recovery paths.
- Documentation must be clear enough to guide implementation.

Acceptance Criteria

- Testing guide can derive checks from this book.
- Acceptance criteria are present for release artifacts.
- Quality risks are recorded before implementation.

Version History

Version	Date	Status	Notes
v1 draft	2026-06-29	Draft	Initial canonical draft for the Mariam Architecture Library.

Book 00-11 - Security Principles

Version: v1 draft

Status: Draft

Document ID: MAL-V00-B011

Purpose

Define the security posture required for Mariam AI Enterprise OS.

Scope

Covers identity, access, data protection, secrets, audit, AI safety, and supply chain awareness.

Principles

- Security is a design constraint, not a final review step.
- Least privilege applies to humans, agents, services, and integrations.
- Sensitive data must have purpose, boundary, and lifecycle.

Operating Rules

- Require explicit permission models for protected actions.
- Never store secrets in documentation, code, prompts, or release artifacts.
- Log security-relevant actions without exposing sensitive content.

Acceptance Criteria

- Future specifications include security considerations.
- Release artifacts contain no secrets.
- AI workflows respect data classification boundaries.

Version History

Version	Date	Status	Notes
v1 draft	2026-06-29	Draft	Initial canonical draft for the Mariam Architecture Library.

Book 00-12 - Engineering Principles

Version: v1 draft

Status: Draft

Document ID: MAL-V00-B012

Purpose

Define engineering behavior expected when Mariam implementation begins.

Scope

Covers architecture discipline, code quality, testing, review, automation, and documentation alignment.

Principles

- Engineering follows documented architecture.
- Small, reversible changes are preferred.
- Automated verification supports human review.

Operating Rules

- Do not implement features without a traceable specification or approved issue.
- Keep module boundaries aligned with blueprint ownership.
- Update documentation when implementation reveals a valid architectural change.

Acceptance Criteria

- Engineering standards can expand from this baseline.
- Pull requests cite relevant documents.
- Implementation work does not outrun approved architecture.

Version History

Version	Date	Status	Notes
v1 draft	2026-06-29	Draft	Initial canonical draft for the Mariam Architecture Library.

Book 00-13 - Evolution Principles

Version: v1 draft

Status: Draft

Document ID: MAL-V00-B013

Purpose

Define how Mariam changes over time without losing coherence.

Scope

Covers versioning, deprecation, migration, compatibility, and learning from production feedback.

Principles

- Evolution should be deliberate and traceable.
- Compatibility matters when users build habits and integrations.
- Deprecation is a managed product experience.

Operating Rules

- Version documents and release artifacts consistently.
- Record migration guidance for breaking changes.
- Preserve historical decisions even when superseded.

Acceptance Criteria

- Future releases can explain what changed and why.
- Deprecated concepts have replacements or retirement rationale.
- Strategic learning feeds back into the library.

Version History

Version	Date	Status	Notes
v1 draft	2026-06-29	Draft	Initial canonical draft for the Mariam Architecture Library.

Book 00-14 - Development Lifecycle

Version: v1 draft

Status: Draft

Document ID: MAL-V00-B014

Purpose

Define the lifecycle from idea to release for Mariam work.

Scope

Covers discovery, documentation, design, implementation, verification, release, operation, and feedback.

Principles

- Documentation leads implementation.
- Acceptance criteria are written before delivery claims.
- Feedback becomes structured improvement, not scattered memory.

Operating Rules

- Start significant work with a document, issue, or specification reference.
- Move from draft to approved status through review.
- Do not release without manifest, notes, and verification evidence.

Acceptance Criteria

- Roadmap items can be staged through lifecycle states.
- Release artifacts demonstrate completed lifecycle steps.
- Contributors understand when work is ready to build.

Version History

Version	Date	Status	Notes
v1 draft	2026-06-29	Draft	Initial canonical draft for the Mariam Architecture Library.

Book 00-15 - Long-Term Strategy

Version: v1 draft

Status: Draft

Document ID: MAL-V00-B015

Purpose

Define the durable strategic posture for Mariam across multiple product eras.

Scope

Covers platform expansion, ecosystem growth, knowledge compounding, enterprise adoption, and responsible AI evolution.

Principles

- Build a platform that compounds through reusable knowledge and governed execution.
- Expand domains through architecture, not ad hoc duplication.
- Keep the system model-independent and organization-aware.

Operating Rules

- Prioritize foundations that unlock multiple future capabilities.
- Avoid vendor, domain, or workflow lock-in unless explicitly justified.
- Review long-term strategy at major release gates.

Acceptance Criteria

- Volume 01 strategic books align with this document.
- Roadmap planning can identify foundation work versus feature work.
- Future expansion remains coherent with the enterprise OS identity.

Version History

Version	Date	Status	Notes
v1 draft	2026-06-29	Draft	Initial canonical draft for the Mariam Architecture Library.

Modular Changeability Principle

Title: Modular Changeability Principle

Version: v130 draft

Status: Draft

Document ID: MAL-V00-B16

Purpose

Define the constitutional rule that every governed object in Mariam is modular, lifecycle-managed, and changeable through approved operations. No capability, model, provider, workflow, agent, dashboard, connector, policy, or runtime object is treated as permanently fixed.

Scope

- Applies to all Mariam Living Enterprise OS Core objects, including runtime objects, DNA packages, models, providers, workflows, agents, services, storage adapters, dashboards, reports, policies, and integrations.
- Defines mandatory lifecycle operations and governance controls.
- Excludes unsafe direct mutation, hidden bypasses, unapproved destructive operations, and changes that skip audit or compatibility checks.

Principle

Every object in Mariam can be changed through a governed lifecycle. The system must support:

- Add
- Edit
- Delete
- Disable
- Enable
- Upgrade
- Replace
- Fork
- Rollback
- Export as DNA
- Import from DNA

Operating Rules

- Add creates a new governed object with owner, version, permissions, dependencies, and acceptance criteria.
- Edit changes an object through versioned revisions, not silent mutation.
- Delete prefers soft delete first; hard delete requires explicit policy, retention, dependency, and audit checks.
- Disable removes an object from active execution while preserving state and evidence.
- Enable reactivates a valid object after compatibility and permission checks.
- Upgrade moves an object to a newer version with migration and rollback planning.

- Replace swaps an implementation or provider behind a stable contract after impact analysis.
- Fork creates a derived object with separate ownership, version lineage, and compatibility metadata.
- Rollback restores a previous approved version or state without erasing audit history.
- Export as DNA packages the object, metadata, dependencies, compatibility rules, and governance evidence.
- Import from DNA validates a package before registration, activation, or execution.

Governance Requirements

- Governance: changes must follow approval policy according to risk, scope, and authority.
- Permission: only authorized roles may perform lifecycle operations.
- Versioning: every meaningful change must produce a traceable version or revision.
- Audit: every lifecycle transition must record actor, time, reason, input, output, decision, and evidence.
- Compatibility: dependency and interface compatibility must be checked before activation.
- Testing: changes require tests appropriate to the object type and risk level.
- Rollback: activation must define rollback or safe-disable behavior before production use.

Acceptance Criteria

- New designs assume objects are changeable unless policy explicitly restricts the operation.
- Documentation and implementation tickets include lifecycle operations for add, edit, delete, disable, enable, upgrade, replace, fork, rollback, export, and import.
- No object can bypass governance, permissions, versioning, audit, compatibility, testing, or rollback.
- DNA export/import is treated as a first-class lifecycle operation.

Traceability

Source	Relationship
MAL-V00-B16	Constitutional principle for modular changeability.
Volume 14 DNA Operating System	DNA managed runtime object model.
Volume 15 Runtime Ecosystem	Runtime/provider lifecycle reference.
Volume 21 Infrastructure	Runtime object governance reference.

Version History

Version	Date	Status	Notes
v130 draft	2026-06-30	Draft	Added modular changeability principle.